



Hypothesis Testing · Worksheet 1

Statistics Chapter 7: Hypothesis Testing. Answer all questions. Show your working clearly. Total: 87 marks. Time: 110 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 7. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- In every test: state H_0 and H_1 using the population parameter, show the probability or critical value you compare with, and write a conclusion **in context** that is not definite (“there is evidence...”).
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A The language of hypothesis testing	16	
B Binomial hypothesis tests	25	
C Testing the mean of a normal distribution	23	
D Testing for correlation	19	
Roar-level challenge	4	
Total	87	

Section A The language of hypothesis testing

16 marks

1. Explain what is meant by (a) the *null hypothesis*, (b) the *alternative hypothesis*. [2]

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2. Explain the difference between a one-tailed test and a two-tailed test. [2]

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3. Explain what is meant by (a) a *critical region*, (b) the *significance level*, (c) the *actual significance level* of a test. [3]

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4. Write down suitable hypotheses, in terms of p , for each situation. [3]

- (a) Amy thinks that a coin is biased towards heads.
- (b) Ben thinks that a die lands on six less often than it should.
- (c) Cara thinks that the proportion of left-handed students, which used to be 0.1, has changed.

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5. A coin is tossed 12 times to test whether it is biased towards heads. The number of heads is X .

(a) Write down suitable hypotheses. [1]

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(b) Using a 5% level of significance, find the critical region for X . [3]

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(c) Write down the actual significance level of the test. [1]

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(d) The coin lands heads 9 times. State the conclusion of the test. [1]

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(Total for Question 5 is 6 marks)



Section B Binomial hypothesis tests

25 marks

6. It is known that 30% of customers buy a drink with their meal. After a promotion, 6 of a random sample of 10 customers buy a drink. [3]
Test, at the 5% level, whether the proportion has increased.

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7. A spinner is claimed to be fair, landing on red with probability 0.5. It lands on red 5 times in 20 spins. [3]
Test, at the 5% level, whether the probability of landing on red is less than 0.5.

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8. A test of $H_0: p = 0.25$ against $H_1: p > 0.25$ is carried out using a sample of 20, at the 5% level. [4]
Find the critical region, and state the actual significance level.

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9. A test of $H_0: p = 0.4$ against $H_1: p \neq 0.4$ is carried out using a sample of 20, at the 5% level. [5]
Find the critical region, with the probability in each tail as close as possible to 2.5%, and state the actual significance level.

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10. In the past, 10% of the laptops sold by a shop were returned. After a change of supplier, the manager believes that the proportion returned has increased. In a random sample of 30 laptops sold, 7 are returned.

(a) Write down suitable hypotheses, defining the parameter you use. [2]

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(b) Test the manager's belief at the 5% level of significance. [4]

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(c) Find the critical region for this test. [3]

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(d) Write down the actual significance level of the test. [1]

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(Total for Question 10 is 10 marks)



Section C Testing the mean of a normal distribution

23 marks

11. $X \sim N(\mu, 4^2)$. A random sample of 16 observations has mean 51.6. [3]

Test $H_0: \mu = 50$ against $H_1: \mu > 50$ at the 5% level.

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12. $X \sim N(\mu, 6^2)$. A random sample of 9 observations has mean 36. [3]

Test $H_0: \mu = 40$ against $H_1: \mu < 40$ at the 1% level.

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13. The masses of bags of flour are normally distributed with standard deviation 5 g. The mean is claimed to be 100 g. A random sample of 25 bags has mean mass 102 g. [4]

Test, at the 5% level, whether the mean mass is different from 100 g.

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14. $X \sim N(\mu, 8^2)$. A test of $H_0: \mu = 40$ against $H_1: \mu > 40$ uses a random sample of size 20 at the 5% level. [4]

Find the critical region for $\bar{X}$.

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15. The lifetimes of a brand of battery are normally distributed with mean 48 hours and standard deviation 3.2 hours. A new production method is introduced, and the manager suspects that the mean lifetime has decreased. A random sample of 40 batteries made by the new method has mean lifetime 47.1 hours. Assume the standard deviation is unchanged.

- (a) State suitable hypotheses and the distribution of $\bar{X}$ under H_0 . [2]

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- (b) Test, at the 5% level of significance, whether there is evidence that the mean lifetime has decreased. [5]

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- (c) Find the p -value of the test, and state the smallest whole-number percentage significance level at which H_0 would be rejected. [2]

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(Total for Question 15 is 9 marks)



Section D Testing for correlation

19 marks

Critical values of the product moment correlation coefficient (one-tailed levels):

n	5%	2.5%	1%
10	0.5494	0.6319	0.7155
12	0.4973	0.5760	0.6581
15	0.4409	0.5140	0.5923
20	0.3783	0.4438	0.5155

16. A student wants to test whether there is positive correlation between two variables. Write down suitable hypotheses, defining any symbols you use. [2]

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17. For a sample of 10 pairs of data, $r = 0.61$. Test, at the 5% level, whether there is evidence of positive correlation in the population. [3]

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18. For a sample of 15 pairs of data, $r = -0.48$. Test, at the 5% level, whether there is evidence of correlation in the population. [3]

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19. For a sample of 20 pairs of data, $r = 0.42$. Test whether there is evidence of positive correlation (a) [3]
at the 1% level, (b) at the 5% level.

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20. Using the large data set, a student finds that the product moment correlation coefficient between daily total sunshine and daily maximum temperature for 12 randomly chosen days at Camborne in 2015 is 0.55.

- (a) Write down suitable hypotheses to test whether there is positive correlation between sunshine [2]
and temperature.

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- (b) Carry out the test at the 5% level of significance, stating your conclusion in context. [3]

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- (c) Explain why the test at the 1% level would lead to a different conclusion. [1]

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- (d) Give two reasons why the student should be cautious about concluding that sunshine and tem- [2]
perature are positively correlated at Camborne throughout the year.

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(Total for Question 20 is 8 marks)



ROAR-LEVEL CHALLENGE

4 marks

21. In the past, 35% of a town's residents used the library. A councillor believes that the proportion has fallen. She plans to ask a random sample of 30 residents. [4]
Using a 1% level of significance, find the critical region for a test of her belief, and state the actual significance level.

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END OF WORKSHEET

Total: 87 marks